Pratyusha Kumar

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EDUCATION

B.S. Computer Science 3.75 GPA
Arizona State University, Tempe, AZ
M.S. Robotics and Autonomous Systems (AI) Dec 2026
Arizona State University, Tempe, AZ

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, MATLAB, MySQL, HTML, CSS, JavaScript, sqlite3, XML
Framework/Libraries: React.JS, Django, Flask, JavaFX, JUnit, Scapy, Pandas, .NET, PyTorch, Tensorflow
Tools: Microsoft Office, Power BI, Microsoft Azure, Git, Jira, Alteryx, Azure DevOps, Taiga, AWS (EC2, S3, SQS, Lambda)

PROFESSIONAL EXPERIENCE

Tech Mahindra, New Delhi, India – Trainee Apr 2023 – Jun 2023

- Practiced using external databases to collect form data and perform basic visual analysis
- Created form prototypes for client applications and upskilled front-end skills by using React

PROJECTS

Finding Potential Joins in Datasets using LLMs Feb 2025 - Dec 2025

- Collaborated with 3 engineers to build a column-level embedding pipeline for structured datasets using Python
- Stored embeddings in an ANN index to enable fast semantic column retrieval from natural-language queries
- Implemented LLM-based re-ranking to refine top similar columns identified by embedding similarity
- Evaluated multiple embedding models, ANN index formats, and LLMs to identify the most accurate and efficient pipeline configuration

ARED - AI Agent Feb 2025 - Dec 2025

- Designed a custom preprocessing workflow that reduced compute usage by 30%, achieving 5.3s avg response times
- Built a multi-model AI architecture leveraging lightweight classification and NER models to gate LLM invocation, improving overall efficiency and end-to-end latency using Python
- Deployed the fully containerized, local AI infrastructure using K3s, hosting model weights and inference servers on-device to support business and customer-facing interactions (function calling, translation, and STT) on an edge system

SASE Website (Webmaster) Dec 2023 - Aug 2024

- Trained and managed a team of 4 developers to update the existing site and its features
- Created a detailed tracking system that tracks student details, emails, attendance with the use of Django and Python
- Made the website more accessible by adding non-technical webpages for various functions and auto-analysis
- Increased efficiency by 15% along with 160 students and an average monthly traffic of 3500+ users using the website to sign in to events and for event information

Country Club Management Jan 2025 - May 2025

- Collaborated with a team of 3 developers to create a country club management using .NET and XML file storage
- Stored user, admin, and staff data using XML and implemented password encryption through SHA256
- Learned how to debug system wide issues and internal server errors and crashes for AWS deployment

Bridge2Africa E-Learning Lab Jan 2023 - Mar 2024

- Led a team of engineers to document and modernize the website for our client (Bridge2Africa) by updating the UI/UX
- Developed a shopping page for the Bridge2Africa along with another engineer by using Shopify, and hosted the website on Digital Ocean at elearning-lab.org
- Learned how to create an efficient work environment for a team, assign tasks and responsibilities that lead to productive outcomes, and clearly communicate with a client

EXTRACURRICULAR ACTIVITIES

Society of Asian Scientists and Engineers, ASU Aug 2024 - Present

Director of Internal Affairs, Vice President

- Act as a leader for a team of 9 students to host social, and professional events to student experience
- Perform duties such as presenting slides, reviewing resumes, conducting mock interviews, and conflict management
- Engage with industry representatives and host industry events for ASU students